



United International University (UIU)

Midterm Examination

IPE 401/IPE3401: Industrial Management/Industrial & Operational Management

Fall Trimester: 2025

Total time: 1 hour 30 minutes

Total marks: 30

There are 4 questions. You must answer any 3 questions

1. (a) A new college graduate has a job with Boeing Aerospace. She plans to borrow \$10,000 now to help buy a car. She has arranged to repay the entire principal plus 8% per year interest after 5 years. Determine the amount she will have to repay and construct a cash flow diagram. [2+2] [CO2]
- (b) Mr. Suman invested some money at a 15.5% interest rate, compounded weekly, for 40 years, and Mr. Sunny invested some money at a 26% interest rate, compounded quarterly, for 30 years, to reach \$ 8 million. Whose investment was higher? And how many years did it take for Sunny to reach 5 million? Show with necessary calculations. [2+2+2] [CO2]
2. (a) Ford Motors has two mutually exclusive investment opportunities. The estimated cash flow for the projects is given below: [2+2+1] [CO1]

Investment opportunity A

Year	0	1	2	3	4	5
Cash Flow	-50,000	15,000	16,500	18,300	27,500	25,700

Investment opportunity B

Year	0	1	2	3	4	5
Cash Flow	-50,000	18,000	12,000	16,200	22,600	19,500

Calculate the **NPV** for both projects and identify which opportunity Ford Motors should invest in? Here, the interest rate (MARR) is **10%** compounded annually.

- (b) If projects A and B are both independent, then which project(s) should be selected based on the **IRR** decision rule? Consider the cost of capital is **9%** annually. [2+2+1] [CO1]

Year	0	1	2	3	4
Project A	-10,000	30,000	40,000	50,000	20,000
Project B	-200,000	65,000	58,000	55,000	37,000

3. (a) Khela Hobe, an online retailer of toys, experiences a monthly demand of 10,000 units for its product *Men-10*. Each time Khela Hobe places an order with the manufacturer, it incurs a fixed ordering cost of \$100. The annual holding cost rate is 20 percent of the unit price. The manufacturer offers the following all-units quantity discount schedule. Determine the **economic order quantity (EOQ)** under this discount schedule and identify **the corresponding unit cost**. [7] [CO2]

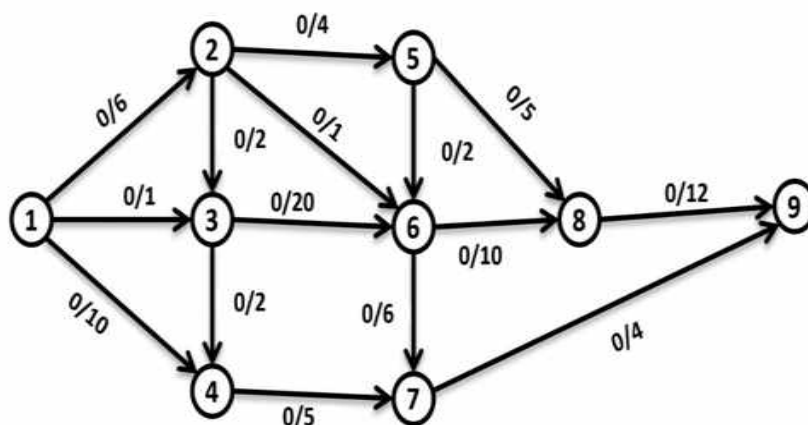
Order Quantity	Unit Price
0-4,999	\$3.00
5,000-9,999	\$2.96
10,000 or more	\$2.92

- (b) Define reorder point (ROP). Draw an approximate **inventory cycle** with proper identifications where order size is 600 units, usage rate is 100 units per day, safety stock is 80 units, and lead time is 2 days. Also show the **reorder point** in the graph. [1+2] [CO1]

4. (a) Find the **critical path**. [5] [CO1]

Task	Predecessors	Duration
A	-	5
B	-	6
C	A	4
D	A	3
E	A	1
F	E	4
G	D, E	14
H	B, C	12
I	G, H	2

- (b) Find the **maximum flow** from the following diagram. [5] [CO2]



CO1-Apply Engineering economics and simple mathematics for Solving project selection problems for choosing the best possible project.

CO2- Analyze various industrial problems by using operation management, technique, operation research technique and cost accounting techniques and solve it.